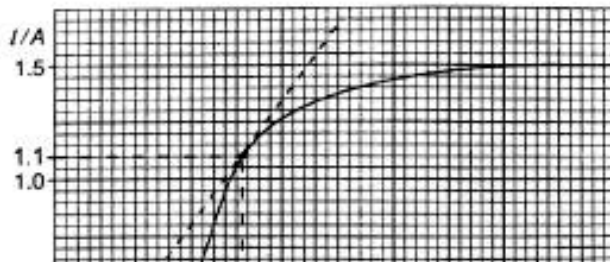


$$\begin{aligned}
 3 \quad (a) \quad (i) \quad R &= \frac{V}{I} \\
 &= \frac{6.0}{1.43} \\
 &= 4.2 \, \Omega
 \end{aligned}$$

- (ii) The minimum value occurs at a point where the ratio of V to I is lowest.
This occurs when $V = 3.3 \, \text{V}$ and $I = 1.1 \, \text{A}$.

$$\text{The resistance } R = \frac{3.3}{1.1} = 3.0 \, \Omega$$



- (b) (i)** The p.d. across the component C is the same as that across the $5.0\ \Omega$ resistor, since both of them are connected in parallel.

$$V = IR$$

$$= (0.85)(5.0)$$

$$= 4.25 = 4.3\text{ V}$$

- (ii)** The total current from the supply is the sum of the current through the $5.0\ \Omega$ resistor (which is 0.85 A) and the current through C (from the graph, when the p.d. across C is 4.3 V , the current through it is 1.3 A).

$$\text{Total current} = 0.85 + 1.3$$

$$= 2.15 = 2.2\text{ A}$$

- (iii)** Terminal p.d. $V = E - Ir$

$$E = V + Ir$$

$$= 4.25 + (2.15)(0.80)$$

$$= 5.97$$

$$= 6.0\text{ V}$$

- (iv)** Energy, $W = Pt = IVt$

$$= (1.3)(4.25)(20 \times 60)$$

$$= 6.6 \times 10^3\text{ J}$$